

Homework 3

Due Date: August 22, 2019 by 5:00pm

MAT 022A Linear Algebra

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Instructions: Write solutions neatly and show all work. Box your solutions, and explain your answers if necessary. Upload a PDF file of your assignment to Gradescope. When uploading to Gradescope, make sure to select which pages correspond to which question.

1. (Hill 2.2.3) Consider a group of five people, P_1 , P_2 , P_3 , P_4 , and P_5 , who have been stationed on a remote island to operate a weather station. The following social interactions have been observed:

P_1 gets along with P_2 , P_3 , and P_4 .

P_2 gets along with P_1 , and P_3 .

P_3 gets along with P_1 , P_2 , and P_4 .

P_4 gets along with P_2 , P_3 , and P_5 .

P_5 gets along with P_4 .

Draw a digraph G describing the situation. Write the adjacency matrix representing G .

2. (Strang 2.4.24) By experiment with $n = 2$ and $n = 3$ predict A^n for these matrices:

$$A_1 = \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix} \quad A_2 = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \quad A_3 = \begin{bmatrix} a & b \\ 0 & 0 \end{bmatrix}$$

3. (Hill 1.4.T.15) Show that $(A - B)^T = A^T - B^T$.
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4. (Hill 1.R.T.1) Show that

a. $\text{tr}(cA) = c \text{tr}(A)$, where c is a real number

b. $\text{tr}(A + B) = \text{tr}(A) + \text{tr}(B)$

c. $\text{tr}(AB) = \text{tr}(BA)$

d. $\text{tr}(A^T) = \text{tr}(A)$

e. $\text{tr}(A^T A) \geq 0$

5. (Strang 2.5.12) If the product $C = AB$ is invertible (A and B are square), then A itself is invertible. Find a formula for A^{-1} that involves C^{-1} and B .

6. (Hill 3.3.3) Evaluate the determinants.

a. $\begin{vmatrix} 1 & 2 & 3 \\ -1 & 5 & 2 \\ 3 & 2 & 0 \end{vmatrix}$

b. $\begin{vmatrix} 4 & -4 & 2 & 1 \\ 1 & 2 & 0 & 3 \\ 2 & 0 & 3 & 4 \\ 0 & -3 & 2 & 1 \end{vmatrix}$

c. $\begin{vmatrix} 4 & -2 & 0 \\ 0 & 2 & 4 \\ -1 & -1 & -3 \end{vmatrix}$

7. (Hill 3.2.8) Let

$$A = \begin{bmatrix} 6 & 2 & 8 \\ -3 & 4 & 1 \\ 4 & -4 & 5 \end{bmatrix}.$$

a. Find $\text{adj } A$.

b. Compute $\det A$.

c. Show that $A(\text{adj } A) = (\text{adj } A)A = \det(A)I$.

8. (Strang 5.3.11) If all entries of A are integers, and $\det A = 1$ or -1 , prove that all entries of A^{-1} are integers. Give a 2 by 2 example with no zero entries.

9. (Hill 3.1.T.6) Is $\det(AB) = \det(BA)$? Justify your answer.

10. (Strang 5.1.1) If a 4 by 4 matrix has $\det A = \frac{1}{2}$, find $\det(2A)$ and $\det(-A)$ and $\det(A^2)$ and $\det(A^{-1})$.